Iris Titanic ML Project Report

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Part A — Iris Dataset Analysis

1. Train–Test Splitting

The Iris dataset was imported from scikit-learn and divided into training and test subsets using a 2/3 to 1/3 ratio. Stratified sampling was applied to preserve class proportions across Setosa, Versicolor, and Virginica. Stratification is essential in classification tasks to ensure balanced representation and reliable performance evaluation.

The dataset consists of 150 observations and four numerical features describing sepal and petal dimensions. The training set was used to fit the model, while the test set was reserved for evaluating generalisation performance.

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2. Feature Scaling

Feature scaling was performed using standardisation (StandardScaler), transforming each feature to have mean 0 and standard deviation 1.

This step is particularly important for K-Nearest Neighbours (KNN), as it is a distance-based algorithm. Without scaling, features with larger numerical ranges would dominate the distance calculations, biasing neighbour selection. The scaler was fitted only on the training data and then applied to the test data to prevent data leakage.

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3. Supervised Learning Using K-Nearest Neighbours

A KNeighborsClassifier was implemented with the following parameters:

- n_neighbors = 7
- metric = “manhattan”
- algorithm = “ball_tree”
- leaf_size = 3

The Manhattan distance metric calculates the sum of absolute differences between feature values. The Ball Tree algorithm improves computational efficiency in multi-dimensional neighbour searches.

The classifier was trained using the scaled training data, and predictions were generated for both training and test sets.

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4. Evaluation on Training Set

Performance was evaluated using a confusion matrix and classification report.

The model achieved a **training accuracy of 0.97**.

Class-level performance:

- Class 0: Precision = 1.00, Recall = 1.00, F1-score = 1.00
- Class 1: Precision = 0.97, Recall = 0.94, F1-score = 0.95
- Class 2: Precision = 0.94, Recall = 0.97, F1-score = 0.96

The macro-average F1-score was **0.97**, indicating strong and consistent classification performance.

Minor misclassifications occurred between Versicolor and Virginica, which is expected due to slight overlap in their feature space. Overall, the model fits the training data effectively without excessive complexity.

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5. Evaluation on Test Set

On the test set, the model achieved an overall **accuracy of 0.94**, demonstrating strong generalisation.

Class-level performance:

- Class 0: Precision = 1.00, Recall = 1.00, F1-score = 1.00
- Class 1: Precision = 0.85, Recall = 1.00, F1-score = 0.92
- Class 2: Precision = 1.00, Recall = 0.82, F1-score = 0.90

The macro-average F1-score was **0.94**.

The small decrease from training accuracy (0.97) to test accuracy (0.94) indicates minimal overfitting. The classifier generalises well to unseen data.

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6. Unsupervised Learning Using K-Means

K-Means clustering was applied to the full dataset with cluster numbers ranging from 2 to 6. The silhouette score was used to evaluate clustering quality.

The highest silhouette score was observed at **k = 3**, which corresponds to the true number of Iris species.

This confirms that the dataset naturally forms three clusters and demonstrates consistency between supervised classification and unsupervised clustering results.

Part B — Titanic Survival Prediction

1. Presentation of the Problem

The objective of this project was to predict whether a passenger survived the Titanic disaster using demographic and travel-related features.

This is a supervised binary classification problem where:

- Target variable: Survived (0 = did not survive, 1 = survived)
- Goal: Develop predictive models capable of generalising to unseen data.

This task represents a real-world risk prediction problem, where historical labelled data is used to model future outcomes.

The analysis followed a structured machine learning workflow:

1. Data exploration
2. Data preparation
3. Model selection
4. Evaluation
5. Conclusion

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2. Data Exploration

The dataset contains 891 observations with features including age, sex, passenger class, fare, and embarkation port.

Exploratory analysis revealed:

- Approximately 38% of passengers survived.
- Females had significantly higher survival rates than males.
- First-class passengers had higher survival probabilities than third-class passengers.
- Age contained missing values.
- Cabin exhibited substantial missingness and was excluded.

Visual analysis confirmed that gender and passenger class were strong predictors of survival, consistent with historical accounts of evacuation priority.

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3. Data Preparation

The following preprocessing steps were implemented:

1. Removed identifier (PassengerId) and high-missingness feature (Cabin).
2. Split data into training (80%) and test (20%) sets using stratification.
3. Applied preprocessing pipelines:
    - Numeric features: median imputation and standard scaling.
    - Categorical features: most frequent imputation and one-hot encoding.

Preprocessing was implemented using ColumnTransformer to ensure consistency and prevent data leakage.

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4. Model Selection

Two supervised learning models were evaluated:

Logistic Regression

Logistic Regression serves as a strong baseline model and offers interpretability. Class balancing was applied to address mild class imbalance.

Random Forest

Random Forest is an ensemble model capable of capturing non-linear feature interactions and complex decision boundaries.

Cross-Validation Results

Five-fold stratified cross-validation was conducted using ROC-AUC as the evaluation metric.

- Logistic Regression CV ROC-AUC: **0.871 ± 0.021**

- Random Forest CV ROC-AUC: **0.880 ± 0.023**

Random Forest achieved slightly higher cross-validation performance, suggesting that non-linear relationships exist within the data.

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5. Evaluation on Test Set

Logistic Regression

- Accuracy: 0.80
- Class 0 (Non-survivors): Precision = 0.84, Recall = 0.85, F1 = 0.84
- Class 1 (Survivors): Precision = 0.75, Recall = 0.74, F1 = 0.74
- Test ROC-AUC: **0.857**

Logistic Regression demonstrates balanced discrimination ability with strong ROC-AUC performance.

Random Forest

- Accuracy: 0.82
- Class 0 (Non-survivors): Precision = 0.81, Recall = 0.91, F1 = 0.86
- Class 1 (Survivors): Precision = 0.82, Recall = 0.67, F1 = 0.74

Random Forest achieved slightly higher overall accuracy but lower recall for survivors compared to Logistic Regression.

Model Comparison

Random Forest achieved marginally higher accuracy and cross-validation ROC-AUC. However, Logistic Regression demonstrated more balanced recall between classes.

This suggests:

- Random Forest better identifies non-survivors.
- Logistic Regression provides more stable class-level discrimination.

Both models demonstrate good generalisation performance, with Random Forest slightly outperforming overall.

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6. Conclusion

This assignment implemented a complete machine learning workflow, including supervised classification, unsupervised clustering, preprocessing pipelines, and model evaluation.

Key findings:

- The Iris dataset demonstrates clear separability, with KNN achieving 94% test accuracy and K-Means confirming three natural clusters.
- In the Titanic dataset, gender and passenger class are strong predictors of survival.
- Random Forest slightly outperformed Logistic Regression, indicating non-linear feature interactions.

Limitations

- Cabin was excluded due to missingness.
- Hyperparameter tuning was limited.
- Historical data may include unobserved confounding variables.

Future Improvements

- Feature engineering (family size, title extraction).
- Systematic hyperparameter optimisation.
- Threshold adjustment for cost-sensitive scenarios.

Overall, the models demonstrate reliable predictive performance and appropriate application of machine learning principles.